INDIA POLLUTION PULSE

Real-Time Air Quality Monitoring Dashboard For India

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Live URL	https://pollutionui-hvckfzekfvfyhqcw.canadacentral-01.azurewebsites.net/

1. Executive Summary

India Pollution Pulse is a live, interactive, and production-ready web dashboard that provides real-time air quality insights across India. It visualizes pollution data from hundreds of government monitoring stations on a dynamic 3D map, automatically calculates the official Air Quality Index (AQI) using CPCB standards, and delivers powerful analytical views.

Designed with a modern dark theme, the dashboard is intuitive for general citizens while offering in-depth analytics for researchers, environmental agencies, and policymakers. Data is fetched hourly through an automated Azure pipeline and displayed instantly.

Data Source	Refresh Rate	Visualization	Deployment
data.gov.in API	Every Hour (Auto)	7 Interactive Tabs	Microsoft Azure

2. Project Objectives

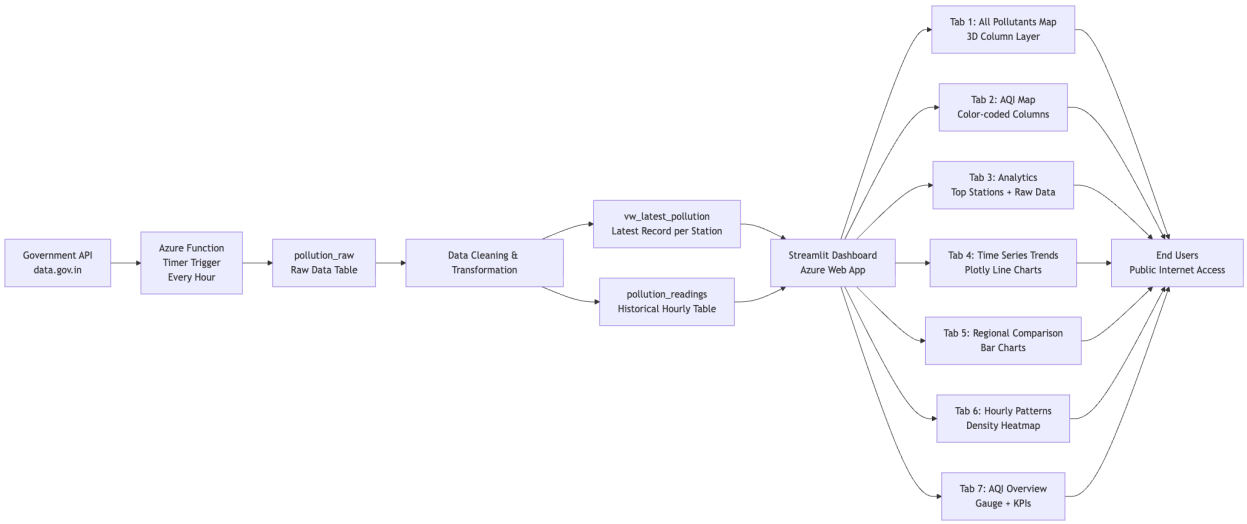
- Deliver a user-friendly, real-time visualization of air pollution across India.
- Automatically compute and display the official CPCB Air Quality Index (AQI).
- Enable flexible filtering by pollutant, region, state, and station.
- Provide multiple analytical perspectives through interactive charts and maps.
- Build a scalable, secure, and fully cloud-native system on Microsoft Azure.

3. System Architecture

3.1 High-Level Architecture

The system follows a three-layer cloud-native architecture:

The detailed data flow shows how raw government data is transformed through the Azure pipeline into the 7-tab interactive dashboard:



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STEP 4: STREAMLIT DASHBOARD (Azure Web App)

Tab 1	Tab 2	Tab 3	Tab 4	Tab 5	Tab 6	Tab 7
All Pollutants Map 3D Column Layer	AQI Map Color-coded Columns	Analytics Top Stations + Raw Data	Time Series Trends / Line Charts	Regional Comparison Bar Charts	Hourly Patterns Density Heatmap	AQI Overview Gauge + KPIs

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END USERS — Public Internet Access

4. Technologies Used

Component	Technology	Purpose
Dashboard Framework	Streamlit (Python)	Interactive web UI
3D Maps	Pydeck	3D column maps of India
Charts & Visualizations	Plotly	Professional charts & gauges
Data Processing	Pandas	Data manipulation
Database Connectivity	pyodbc + ODBC Driver 18	Secure SQL connection
Database	Azure SQL Database	Data storage

ETL Pipeline	Azure Functions (Timer Trigger)	Hourly data ingestion
Hosting & Deployment	Azure App Service (Linux)	Production web hosting
Programming Language	Python 3.11	Core development

5. Database Design

The database consists of three core tables/views, all properly indexed for high performance:

Table / View	Description
`pollution_raw`	Stores every raw record received from the Government API. Acts as the primary landing zone for ingested data.
`vw_latest_pollution`	Optimized SQL view returning only the most recent reading per monitoring station. Used by live map tabs.
`pollution_readings`	Historical hourly table storing all processed readings over time. Used for trend analysis and time series charts.

6. AQI Calculation Formula (CPCB Standard)

India Pollution Pulse implements the official CPCB (Central Pollution Control Board) AQI calculation methodology. The "Worst Pollutant" method is used — the final AQI for a station equals the highest sub-index across all pollutants.

6.1 AQI Formula

For each pollutant, the sub-index is calculated using linear interpolation:

```
AQI = ( (AQI_High - AQI_Low) / (C_High - C_Low) ) × (C - C_Low) + AQI_Low

Where:
  C          =  Measured concentration of the pollutant
  C_Low      =  Lower bound of the concentration range
  C_High     =  Upper bound of the concentration range
  AQI_Low    =  AQI value at C_Low breakpoint
  AQI_High   =  AQI value at C_High breakpoint
```

6.2 CPCB AQI Breakpoints

AQI Range	Category	PM2.5 (µg/m³)	PM10 (µg/m³)	NO_2 (µg/m³)	SO_2 (µg/m³)
0–50	Good	0–30	0–50	0–40	0–40
51–100	Satisfactory	31–60	51–100	41–80	41–80
101–200	Moderate	61–90	101–250	81–180	81–380
201–300	Poor	91–120	251–350	181–280	381–800
301–400	Very Poor	121–250	351–430	281–400	801–1600
401–500	Severe / Hazardous	250+	430+	400+	1600+

6.3 Python Implementation

The AQI calculation function in the codebase:

```
def calculate_station_aqi(station_data):
    aqi_values = []
    breakpoints = {
        'PM2.5': [(0,30,0,50),(31,60,51,100),(61,90,101,200),...],
        'PM10':  [(0,50,0,50),(51,100,51,100),(101,250,101,200),...],
        'NO2':   [(0,40,0,50),(41,80,51,100),(81,180,101,200),...],
        'SO2':   [(0,40,0,50),(41,80,51,100),(81,380,101,200),...],
    }
    for pollutant, bp_list in breakpoints.items():
        C = station_data.get(pollutant)  # measured concentration
        if C is None or pd.isna(C):      # skip if no reading
            continue
        for (C_Low, C_High, AQI_Low, AQI_High) in bp_list:
            if C_Low <= C <= C_High:
                sub_index = ((AQI_High - AQI_Low) /
                             (C_High  - C_Low)) * (C - C_Low) + AQI_Low
                aqi_values.append(round(sub_index))
                break
    return max(aqi_values) if aqi_values else None  # Worst Pollutant Method
```

7. Dashboard Features (Tab-by-Tab)

The dashboard contains 7 interactive tabs, each with independent filters:

Tab 1: All Pollutants Map	3D column visualization where column height represents maximum pollution value across all pollutants. Allows filtering by individual pollutant (PM2.5, PM10, NO_2, SO_2, CO, O_3).
Tab 2: AQI Map	Color-coded 3D columns mapped to AQI categories (Green = Good → Dark Red = Hazardous). Provides instant visual identification of heavily polluted zones.

Tab 3: Analytics	Top 15 most polluted stations displayed in ranked view, along with a full raw data table. Supports region, state, and station filtering.
Tab 4: Time Series Trends	Interactive Plotly line charts showing hourly and daily pollution trends per station. Ideal for tracking patterns over time.
Tab 5: Regional Comparison	Bar charts comparing pollution levels by region and top 10 states. Useful for inter-regional benchmarking.
Tab 6: Hourly & Daily Patterns	Density heatmap showing average pollution levels by hour-of-day and day-of-week, plus daily trend lines.
Tab 7: AQI Overview & KPIs	National AQI gauge displaying the overall air quality level, plus key metrics including most polluted city, cleanest city, and national average.

8. Special Features

- **Real-time AQI calculation using official CPCB breakpoints and the Worst Pollutant method.**
- Modern dark professional theme for improved readability and visual impact.
- Independent smart filters on every tab — state, region, station, and pollutant.
- Automatic data refresh every 5 minutes in the Streamlit UI.
- Clear India map outline rendered using GeoJSON boundary data.
- Fully modular, component-based code structure for easy maintenance.

9. Key Technical Highlights

9.1 Secure Database Connection

The database connection string is never hardcoded. It is stored securely in Azure Application Settings and accessed at runtime via environment variables:

```
import os
conn_str = os.environ['CONNECTION_STRING']
conn = pyodbc.connect(conn_str)
```

9.2 3D Map Rendering

Pydeck ColumnLayer is used with dynamic elevation (proportional to pollution value) and fill color (mapped to AQI category). The India GeoJSON boundary is overlaid for geographic context.

9.3 Performance Optimization

- @st.cache_data decorator applied to all data fetch functions.
- Optimized SQL queries using the vw_latest_pollution view.
- Indexes applied on station_id, timestamp, and state columns.

9.4 Modular Architecture

All 7 tabs are implemented as separate, reusable Python modules inside a components/ folder. The main app.py only handles routing and sidebar filters.

10. How to Run the Project Locally

1. Clone the repository from the source control system.
2. Create and activate a Python virtual environment.
3. Install all dependencies: pip install -r requirements.txt
4. Set the CONNECTION_STRING environment variable with your Azure SQL connection string.
5. Launch the dashboard: streamlit run app.py

```
# Example: set environment variable (Linux/Mac)
export CONNECTION_STRING='Driver={ODBC Driver 18 for SQL Server};Server=...'

# Run the app
streamlit run app.py
```

11. Deployment on Azure

Setting	Value
Hosting	Azure App Service (Linux)
Startup Command	python -m streamlit run app.py --server.port 8000 --server.address 0.0.0.0
SCM_DO_BUILD_DURING_DEPLOYMENT	true
CONNECTION_STRING	Stored securely in Application Settings
Auto Refresh	Every 5 minutes via Streamlit rerun

12. Challenges Faced & Solutions

Challenge	Solution Implemented
ODBC Driver not available on Azure	Used Microsoft ODBC Driver 18 with correct Linux configuration
Modular code deployment issues	Enabled SCM_DO_BUILD_DURING_DEPLOYMENT=true in App Settings
Hardcoded credentials risk	Moved all secrets to Azure Application Settings
Slow map performance	Implemented @st.cache_data and optimized SQL queries
AQI calculation errors	Fixed CPCB breakpoints and added proper NaN handling

13. Future Enhancements (Phase 2)

- Historical date range selector with interactive time slider.
- Export to Excel / CSV functionality.
- Email and SMS alerts when AQI exceeds threshold levels.
- User authentication and personalized dashboards.
- Weather data integration for correlation analysis.
- Mobile-responsive UI improvements.

14. Conclusion

India Pollution Pulse is a complete, production-ready real-time air quality monitoring dashboard. It successfully integrates government data sources, cloud infrastructure, and modern visualization techniques to deliver clear, actionable, and visually appealing insights into India's air quality.

This project demonstrates end-to-end full-stack development capabilities — from data ingestion and ETL pipeline design to interactive dashboard development and secure cloud deployment.

Live Dashboard
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